

Set Up Network for Raspberry Pi

Learn to set up wired and wireless networks for your Raspberry Pi



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After having your Raspberry Pi up and running, you will need an Internet connection to develop interesting applications on it. You can easily connect your Raspberry Pi to the Internet using a 10/100 Ethernet or a wireless adaptor. Here is how you can open www.google.com webpage using the in-built browsers.

Wired LAN

Insert the 10/100 Ethernet cord into the network port of the Raspberry Pi. Open a terminal window on your Raspberry Pi and type 'ifconfig.' If your network works in dynamic host control protocol (DHCP) mode (which is supported by most broadband networks), it will assign the IP address automatically and the IP will be seen on the screen as shown in Fig. 1.

But if the network demands a static IP address, get ready for another interesting job that awaits you.

Open the terminal window again and type 'sudo nano/etc/network/interfaces.' Nano is a light-weight text editor. It is sleek, fast and efficient and can open the network interface file for you as shown in Fig. 2.

Now try and locate 'iface eth0 inet dhcp.'

Here 'dhcp' means that an automatic IP address will be assigned by the network to your Raspberry Pi. First, replace the word 'dhcp' with 'static' and then below it write the following lines (first line you already have):

```
iface eth0 inet static
[tab] address 10.0.73.220
[tab] netmask 255.255.255.0
[tab] gateway 10.0.72.5
```

'tab' is used here just to create a margin of one character at the leftmost

space, which the network protocol invariably skips while parsing. Insert the IP address, net mask and gateway that you want. Do not edit lines any

further. Save it (by 'ctrl+s'), exit (by 'ctrl+x') and restart the network by:

```
sudo /etc/init.d/networking restart
```

Now if everything goes well, the

```
pi@raspberrypi ~$ ifconfig
eth0:  Link encap:Ethernet  HWaddr b8:27:eb:d3:ee:e3
       inet addr:192.168.2.197  Bcast:192.168.2.255  Mask:255.255.255.0
       UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
       RX packets:1169 errors:0 dropped:2 overruns:0 frame:0
       TX packets:353 errors:0 dropped:0 overruns:0 carrier:0
       collisions:0 txqueuelen:1000
       RX bytes:77968 (76.1 KiB)  TX bytes:369281 (360.6 KiB)
```

Fig. 1: ifconfig command and IP allotted

```
pi@raspberrypi ~$ nano /etc/network/interfaces
GNU nano 2.2.6 /etc/network/interfaces
auto lo
iface lo inet loopback
iface eth0 inet dhcp

allow-hotplug wlan0
iface wlan0 inet manual
wpa-conf /etc/wpa-supplicant/wpa_supplicant.conf
iface default inet dhcp
```

Fig. 2: Network interfaces

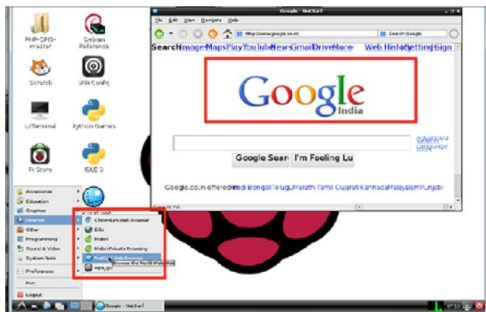


Fig. 3: NetSurf Web browser with google webpage